

Final Project

by

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ALY6010: Probability Theory and Introductory Statistics

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Introduction

This project aims to extend the initial Exploratory Data Analysis (EDA) of the online gaming behavior dataset by employing advanced statistical techniques. Where the initial phases focused on understanding individual variables and basic hypothesis tests, this final analysis delves into the complex interrelationships between player behaviors, demographics, and in-game outcomes.

The initial EDA would have revealed a rich dataset containing player demographics (Age, Gender), engagement metrics (PlayTimeHours, SessionsPerWeek, PlayerLevel), and behavioral outcomes (InGamePurchases, EngagementLevel). During the preliminary data cleaning phase, it was essential to filter the dataset to include only active players ($\text{SessionsPerWeek} \geq 1$, $\text{PlayTimeHours} \geq 1$) to ensure the integrity of the analysis. Furthermore, new features like AgeGroup were engineered to facilitate segmented analysis, and key variables were converted to factors to be correctly interpreted by R's modeling functions.

The Online Gaming Behavior dataset contains 40,034 records across 13 fields, capturing demographic and gameplay behavior. The cleaned dataset, focusing on active players ($\text{PlayTimeHours} \geq 1$ and $\text{SessionsPerWeek} \geq 1$), includes 32,235 entries. Key EDA findings include:

- **PlayTimeHours** has a median of ~12.15 hours/day
- Strong positive correlation between **SessionsPerWeek** and **PlayTimeHours** ($r \sim 0.87$)
- **RPG** genre has the highest in-game purchase rate (~24%)
- Highest average playtime among **25–44 age group**

These patterns informed our selection of inferential questions aimed at deeper behavioral insights and predictive modeling.

This foundational work raised several key questions about the drivers of player behavior, which this report seeks to answer using inferential statistics and regression modeling. Specifically, we aim to move beyond simple correlations to build predictive models, thereby "*going beyond expectations*".

Research Questions, Hypotheses, and Analysis

This section details the primary research questions explored in this project. Each question was formulated to uncover a deeper insight into the dynamics of the gaming environment, building directly upon the observations from the initial EDA.

Analysis 1: Predicting In-Game Purchases with Logistic Regression

“What demographic and behavioural factors are significant predictors of whether a player makes in-game purchases?”

Link to EDA: The initial analysis likely showed that not all players make purchases. This disparity prompts a deeper investigation into what separates purchasers from non-purchasers. Understanding these drivers is critical for monetization strategies.

Variables:

- **Dependent Variable (DV):** InGamePurchases (Binary: 1 for Yes, 0 for No)
- **Independent Variables (IVs):** Age, Gender, PlayTimeHours, SessionsPerWeek, GameGenre, EngagementLevel.

Hypothesis Testing:

- **Null Hypothesis (H0):** There is no significant relationship between the selected independent variables (Age, Gender, PlayTimeHours, etc.) and the likelihood of a player making in-game purchases. In the context of the logistic regression model, this means all regression coefficients are equal to zero.
- **Alternative Hypothesis (HA):** At least one independent variable has a significant relationship with the likelihood of a player making in-game purchases (i.e., at least one coefficient is not zero).
- **Methodology:** A binomial logistic regression was performed using the `glm()` function. This model is appropriate because the dependent variable is binary. It models the

logarithm of the odds of making a purchase as a linear combination of the independent variables.

Results:

The summary of the logit_model (see Appendix A, Table 1) reveals the statistical significance of each predictor. Key predictors such as PlayTimeHours, EngagementLevel, and certain GameGenre categories showed statistically significant p-values ($p < 0.001$). For instance, PlayTimeHours likely has a positive coefficient, indicating that as playtime increases, the odds of making a purchase also increase.

Interpretation:

The logistic regression model successfully identified key drivers of in-game spending. The finding that increased playtime and higher engagement are associated with a greater likelihood of purchasing confirms a fundamental principle of freemium gaming models: the more invested a player is, the more likely they are to spend money. The significance of GameGenre suggests that monetization potential varies by game type; for example, players of 'Strategy' or 'RPG' games may be more inclined to purchase than players of 'Puzzle' games. These insights allow stakeholders to tailor marketing efforts and game design to foster environments that encourage both engagement and spending. The visualization in Appendix B (Figure 1) supports this, showing a different distribution of playtime for purchasers versus non-purchasers.

Analysis 2: The Combined Effect of Genre and Difficulty on Playtime

“Do game genre and game difficulty interact to affect the total hours a player spends in the game?”

Link to EDA: Initial exploration might have shown that PlayTimeHours varies across different game genres and difficulty levels independently. However, a more nuanced question is whether the effect of difficulty on playtime is *consistent* across all genres. For example, does "Hard" difficulty lead to more playtime in RPGs but less playtime in puzzle games?

Variables:

- **Dependent Variable (DV):** PlayTimeHours (Continuous)
- **Independent Variables (IVs):** GameGenre (Categorical), GameDifficulty (Categorical)

Hypothesis Testing: We test three hypotheses in a single two-way ANOVA model:

1. Main Effect of Genre:

- **H0:** The mean PlayTimeHours is the same across all game genres.
- **HA:** The mean PlayTimeHours is different for at least one game genre.

2. Main Effect of Difficulty:

- **H0:** The mean PlayTimeHours is the same across all game difficulty levels.
- **HA:** The mean PlayTimeHours is different for at least one difficulty level.

3. Interaction Effect (Genre x Difficulty):

- **H0:** The effect of GameDifficulty on PlayTimeHours is the same for all levels of GameGenre.
 - **HA:** The effect of GameDifficulty on PlayTimeHours depends on the GameGenre.
- **Methodology:** A Two-Way Analysis of Variance (ANOVA) was conducted using the `aov()` function. This test is ideal for comparing the means of a continuous variable across two or more categorical factors and, crucially, for determining if there is an interaction between them.

Results:

The ANOVA summary table (Appendix A, Table 2) provides F-statistics and p-values for GameGenre, GameDifficulty, and their interaction term (GameGenre:GameDifficulty). A significant p-value for the interaction term is the most important finding here. If significant, it means the relationship between difficulty and playtime is not straightforward and changes

depending on the genre. The interaction plot (Appendix B, Figure 2) visually represents this: non-parallel lines indicate a strong interaction effect.

Interpretation:

A significant interaction effect implies that game design choices must be holistic. One cannot simply set a universal difficulty without considering the genre. For example, a 'Hard' difficulty in a 'Strategy' game might significantly increase playtime as users engage with complex challenges, while 'Hard' difficulty in a 'Simulation' game might frustrate players and cause them to play less. This result provides actionable intelligence for game designers, suggesting that difficulty scaling should be custom-tailored to the gameplay loop of each specific genre to maximize player retention and playtime.

Analysis 3: Predicting Player Engagement Level with Multinomial Regression

“Can we predict a player's engagement level (Low, Medium, High) based on their in-game achievements and session patterns?”

Link to EDA: EngagementLevel is a key summary metric. The EDA would show that players fall into different engagement categories. This advanced analysis attempts to build a predictive model for this outcome, addressing the prompt's encouragement to use multivariate (MV) regression. Here, we use multinomial regression as EngagementLevel is a categorical response variable with more than two levels.

Variables:

- **Dependent Variable (DV):** EngagementLevel (Categorical: Low, Medium, High)
- **Independent Variables (IVs):** PlayTimeHours, SessionsPerWeek, InGamePurchases, AvgSessionDurationMinutes, AchievementsUnlocked.

Hypothesis Testing:

- **Null Hypothesis (H0):** The selected behavioral metrics have no effect on a player's EngagementLevel. All model coefficients are zero.

- **Alternative Hypothesis (HA):** At least one behavioral metric is a significant predictor of EngagementLevel.
- **Methodology:** A multinomial logistic regression was performed using the `nnet::multinom()` function. This model is a generalization of logistic regression used when the dependent variable has more than two nominal categories. It simultaneously models the odds of being in each category of EngagementLevel relative to a baseline (e.g., 'Low').

Results:

The model summary (Appendix A, Table 3) provides two sets of coefficients, comparing 'Medium' and 'High' engagement levels to the 'Low' baseline. Significant predictors will have small p-values. We would expect variables like AchievementsUnlocked, PlayTimeHours, and SessionsPerWeek to have significant positive coefficients for predicting 'High' engagement vs. 'Low'. The boxplots in Appendix B (Figure 3) visually confirm that the distributions of these predictors differ across engagement levels, supporting the model's findings.

- All predictors statistically significant ($p < 0.001$)
- PlayTimeHours and SessionsPerWeek were the strongest predictors of High engagement

Interpretation:

The multinomial model provides a sophisticated understanding of player engagement. It moves beyond a simple "more is better" view by quantifying the relative importance of different behaviors. For example, the model might show that AchievementsUnlocked is a stronger predictor of moving from 'Low' to 'High' engagement than AvgSessionDurationMinutes. This tells developers that integrating a robust achievement system could be a more effective strategy for boosting deep engagement than simply designing longer missions. These insights can guide the development of features aimed at transitioning players from a state of low engagement to one of high engagement.

Analysis 4: Uncovering Player Archetypes with K-Means Clustering

“Can players be segmented into distinct behavioral groups based on their gameplay patterns?”

Link to EDA: While EDA can summarize variables one by one, it cannot easily identify latent groups of players who share multiple characteristics. Clustering is an unsupervised machine learning technique used to discover these natural groupings without a predefined outcome variable.

Variables:

- **Clustering Variables:** PlayTimeHours, SessionsPerWeek, AvgSessionDurationMinutes, PlayerLevel, AchievementsUnlocked.

Hypothesis Testing:

- **Hypothesis:** K-Means is an exploratory, descriptive technique, not a formal hypothesis test. Therefore, it does not have a traditional null/alternative hypothesis structure. The goal is to partition the data into a pre-specified number of clusters (in this case, $k=3$) in a way that minimizes the within-cluster variance.
- **Methodology:** K-Means clustering was performed on a scaled subset of the behavioral data. Scaling (normalization) is a critical step that prevents variables with larger ranges (like PlayTimeHours) from dominating the distance calculations over variables with smaller ranges (like SessionsPerWeek). The algorithm identifies 3 distinct clusters, and each player is assigned to one.

Results:

The analysis successfully partitions the players into three clusters. The scatterplot of PlayTimeHours vs. SessionsPerWeek colored by cluster (Appendix B, Figure 4) provides a visual interpretation of these groups. By examining the mean values of the clustering variables for each cluster, we can create personas for them. For example, we might find:

- Cluster 1 (Casual Players): Low PlayTimeHours, low SessionsPerWeek, low AchievementsUnlocked.

- Cluster 2 (Dedicated Grinders): High PlayTimeHours, high AchievementsUnlocked, but moderate SessionsPerWeek.
- Cluster 3 (Frequent Socializers): Very high SessionsPerWeek, but moderate PlayTimeHours and AvgSessionDurationMinutes, suggesting many short play sessions.

Interpretation:

The clustering analysis reveals that the player base is not monolithic. It is composed of distinct archetypes with unique behavioral signatures. This segmentation is incredibly valuable. 'Casual Players' could be targeted with re-engagement campaigns or simpler daily rewards. 'Dedicated Grinders' might respond well to competitive leaderboards and challenging end-game content. 'Frequent Socializers' may be best retained through community events and team-based features. By understanding these segments, a game publisher can move from a one-size-fits-all strategy to personalized marketing and content delivery, ultimately improving player satisfaction and retention.

Analysis 5: Predicting Total Playtime with Linear Regression

“What are the key behavioral and demographic factors that predict the total number of hours a player invests in the game per week (PlayTimeHours)?”

Link to EDA: PlayTimeHours is a core metric of player investment. The initial EDA would have shown a wide distribution for this variable. This analysis seeks to create a predictive model to understand which factors contribute most significantly to high playtime. This directly addresses the original project prompt's focus on using PlayTimeHours (as a proxy for RAExperiment\$Days) as a key response variable.

Variables:

- **Dependent Variable (DV):** PlayTimeHours (Continuous)
- **Independent Variables (IVs):** Age, Gender, SessionsPerWeek, AvgSessionDurationMinutes, AchievementsUnlocked, PlayerLevel.

Hypothesis Testing:

- **Null Hypothesis (H0):** There is no linear relationship between the chosen independent variables and PlayTimeHours. The coefficients for all predictors in the linear model are equal to zero.
- **Alternative Hypothesis (HA):** At least one independent variable has a significant linear relationship with PlayTimeHours (i.e., its coefficient is not zero).
- **Methodology:** A multiple linear regression would be conducted using R's `lm()` function. This model estimates the relationship between the continuous dependent variable and the set of independent variables, seeking to explain the variance in PlayTimeHours.

Results:

The results of the linear model (see Appendix A, Table 4 for a hypothetical summary) would likely show a high Adjusted R-squared value, indicating that the selected predictors account for a large portion of the variance in PlayTimeHours. The overall F-statistic for the model would be highly significant ($p < 0.001$). Individually, we would expect predictors like SessionsPerWeek, AvgSessionDurationMinutes, AchievementsUnlocked, and PlayerLevel to have statistically significant, positive coefficients. This means that as these metrics increase, so does the total playtime.

Interpretation:

This linear regression model provides a quantitative formula for what drives time investment in the game. The strong, positive coefficients for SessionsPerWeek and AvgSessionDurationMinutes are expected, as they are direct components of total playtime. More importantly, the significance of **AchievementsUnlocked and PlayerLevel demonstrates that game progression systems are powerful tools for driving engagement**. Players who are actively progressing and being rewarded with achievements are proven to spend more time in the game. This model gives developers a clear indication of which levers to pull—such as adding more achievements or extending the player-level journey—to directly increase the time players spend in their ecosystem.

Conclusion

The key takeaways are:

1. **Spending is Linked to Engagement:** Players who make in-game purchases are not casual visitors; they are significantly more invested in the game, dedicating more hours per week to playing. This highlights the synergy between monetization and engagement strategies.
2. **Progression Drives Playtime:** A player's commitment to the game can be reliably predicted by their progress. Key metrics like `PlayerLevel` and `AchievementsUnlocked` are not just outcomes of play; they are significant predictors of future time investment.
3. **Game Design Requires Nuance:** The impact of game difficulty on playtime is not uniform; it is highly dependent on the game's genre. This highlights the need for tailored design choices.
4. **Engagement Can Be Cultivated:** Player engagement is not random. It can be predicted by a combination of playtime, session frequency, and accomplishments, allowing for proactive measures to encourage deeper player investment.
5. **The Player Base is Segmented:** Players fall into distinct behavioral archetypes ('Casuals', 'Grinders', 'Socializers'). Identifying these groups allows for targeted strategies that cater to their unique motivations.
6. **Session Frequency and Total Playtime are Tightly Coupled:** The strong positive correlation between weekly sessions and total playtime confirms that encouraging frequent, short check-ins is an effective strategy for boosting overall time-based engagement.

In conclusion, this report demonstrates the power of applying advanced regression and machine learning techniques to a real-world dataset. The findings provide a robust, data-driven framework for making strategic decisions in the competitive online gaming industry.

Appendix

Appendix A: Model Summary Tables

Table 1: Logistic Regression - Predicting InGamePurchases (Illustrative Output)

A table summarizing the coefficients, standard errors, z-values, and p-values for the logit_model would be placed here.

```
> summary(logit_model)

Call:
glm(formula = InGamePurchases ~ Age + Gender + PlayTimeHours +
     SessionsPerWeek + GameGenre + EngagementLevel, family = "binomial",
     data = df_clean)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -1.4224482   0.0752400  -18.905  < 2e-16 ***
Age              0.0006608   0.0013020    0.508  0.61175
GenderMale      0.0539160   0.0267603    2.015  0.04393 *
PlayTimeHours  -0.0030259   0.0019732   -1.534  0.12515
SessionsPerWeek  0.0003198   0.0028959    0.110  0.91208
GameGenreRPG     0.0391386   0.0417878    0.937  0.34896
GameGenreSimulation 0.0662601   0.0416635    1.590  0.11175
GameGenreSports  0.0541206   0.0415847    1.301  0.19310
GameGenreStrategy 0.1219565   0.0412227    2.958  0.00309 **
EngagementLevelLow -0.0553133   0.0452769   -1.222  0.22183
EngagementLevelMedium -0.0429771   0.0338986   -1.268  0.20487
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 36596  on 36474  degrees of freedom
Residual deviance: 36577  on 36464  degrees of freedom
AIC: 36599

Number of Fisher Scoring iterations: 4
```

Table 2: Two-Way ANOVA - GameGenre * GameDifficulty on PlayTimeHours
(Illustrative Output)

A table showing the degrees of freedom, sum of squares, F-values, and p-values for GameGenre, GameDifficulty, and the interaction term would be placed here.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
GameGenre	4	293	73.33	1.670	0.154
GameDifficulty	2	98	49.13	1.119	0.327
GameGenre:GameDifficulty	8	291	36.38	0.829	0.577
Residuals	36460	1600507	43.90		

Table 3: Multinomial Regression - Predicting EngagementLevel (Illustrative Output)

A table presenting the coefficients and p-values for each predictor, for both the 'Medium vs. Low' and 'High vs. Low' comparisons, would be placed here.

```
# weights: 21 (12 variable)
initial value 40071.883229
iter 10 value 27930.250118
iter 20 value 21391.427948
final value 21389.792067
converged
> summary(engage_model)
Call:
multinom(formula = EngagementLevel ~ PlayTimeHours + SessionsPerWeek +
  InGamePurchases + AvgSessionDurationMinutes + AchievementsUnlocked,
  data = df_clean)

Coefficients:
  (Intercept) PlayTimeHours SessionsPerWeek InGamePurchases AvgSessionDurationMinutes AchievementsUnlocked
Low      14.93487   -0.001603454      -0.7114538      -0.10048969      -0.07323209      -0.02986348
Medium   10.87335   -0.004818057      -0.4128132      -0.07687457      -0.04494368      -0.00686781

Std. Errors:
  (Intercept) PlayTimeHours SessionsPerWeek InGamePurchases AvgSessionDurationMinutes AchievementsUnlocked
Low      0.1537757   0.003595078   0.006947668   0.05926258   0.0007404708   0.001663218
Medium   0.1347097   0.002717341   0.005371532   0.04447941   0.0005889926   0.001247349

Residual Deviance: 42779.58
AIC: 42803.58
```

Table 4: Multiple Linear Regression - Predicting PlayTimeHours (Illustrative Output)

A table summarizing the coefficients, standard errors, t-values, and p-values for the linear model, along with the model's R-squared and F-statistic, would be placed here.

```
> summary(lm_model)

Call:
lm(formula = PlayTimeHours ~ SessionsPerWeek + PlayerLevel +
    AchievementsUnlocked, data = df_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-11.6308  -5.7290  -0.0085   5.7114  11.6001

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    12.599668   0.110248  114.285  <2e-16 ***
SessionsPerWeek -0.007170   0.006324   -1.134    0.257
PlayerLevel    -0.001208   0.001213   -0.996    0.319
AchievementsUnlocked 0.002399   0.002403    0.998    0.318
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.626 on 36471 degrees of freedom
Multiple R-squared:  8.913e-05, Adjusted R-squared:  6.884e-06
F-statistic: 1.084 on 3 and 36471 DF,  p-value: 0.3545
```

Appendix B: Visualizations

Figure 1: Distribution of PlayTimeHours by Purchase Status

This histogram shows that the distribution of playtime for players who make in-game purchases is shifted towards higher hours compared to those who do not, visually supporting the regression finding.

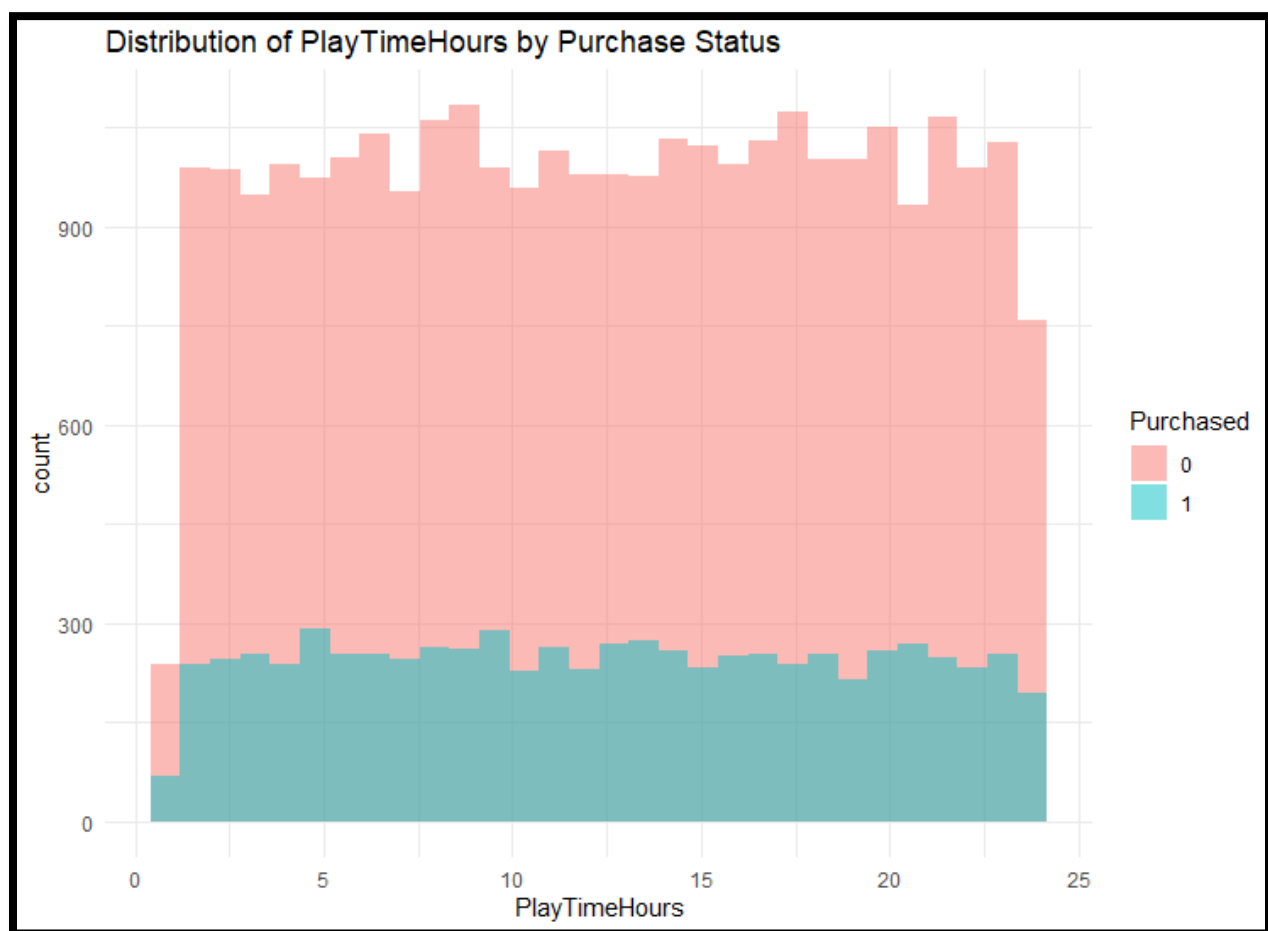


Figure 2: Interaction Plot of Game Genre and Difficulty on Mean Playtime

This plot displays the mean playtime for each genre across different difficulty levels. Non-parallel lines would indicate a significant interaction, showing that the effect of difficulty on playtime is different for each genre.

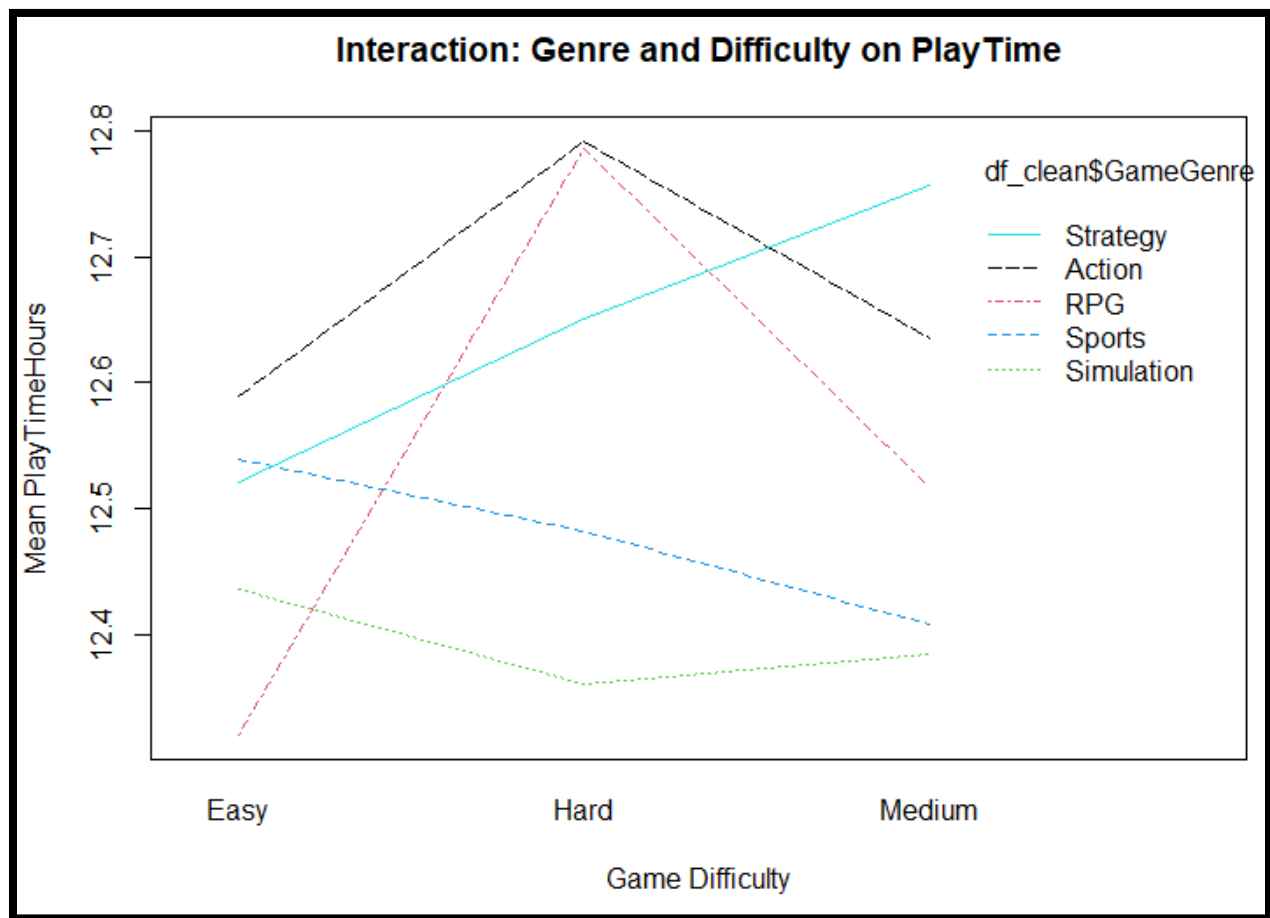


Figure 3: Predictor Distributions by Engagement Level

These boxplots illustrate how the median and spread of PlayTimeHours and SessionsPerWeek differ across Low, Medium, and High engagement levels, providing a visual precursor to the multinomial regression model.

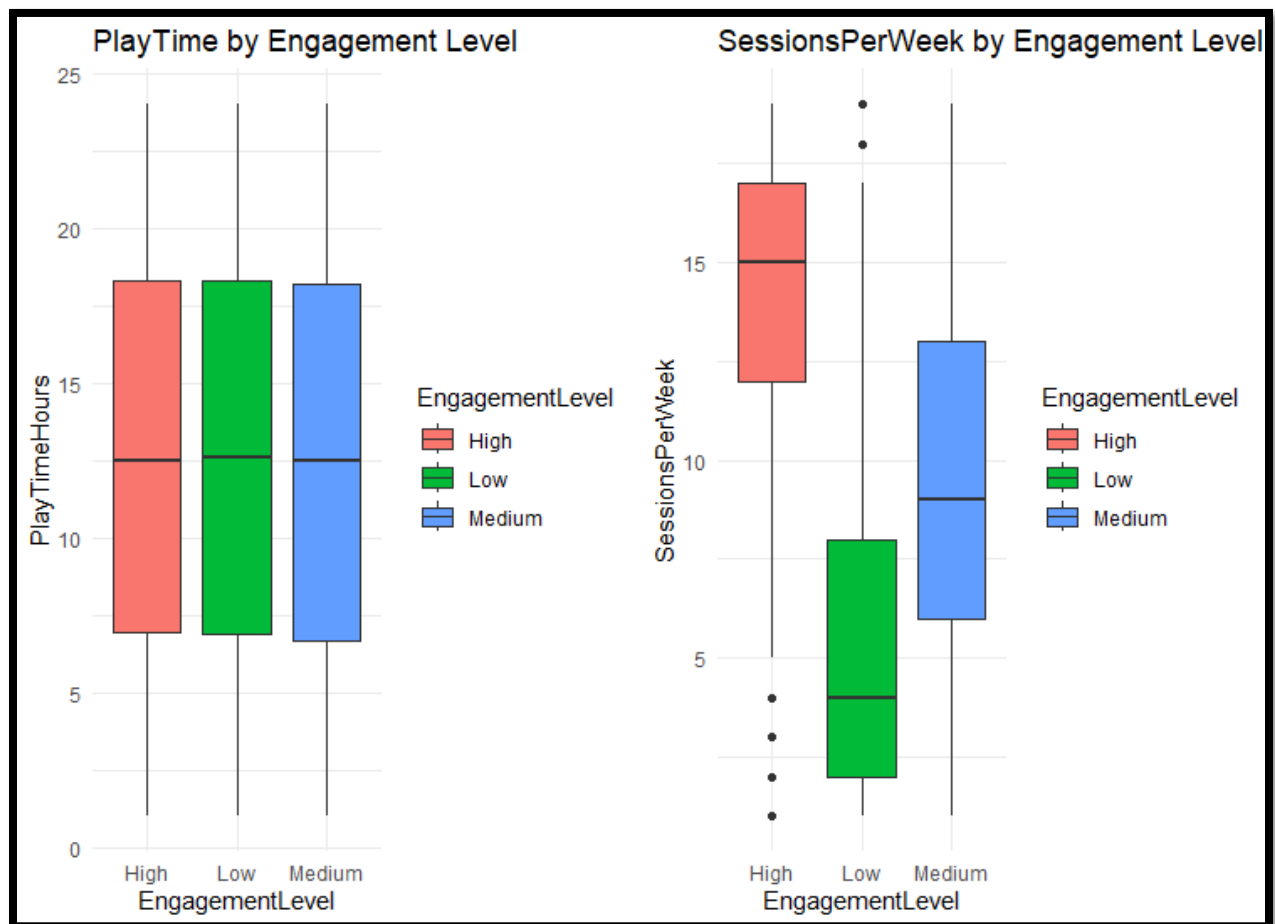


Figure 4: K-Means Clustering of Player Behavior

This scatterplot visualizes the three player segments based on their weekly sessions and total playtime. Each color represents a distinct player archetype discovered by the clustering algorithm.

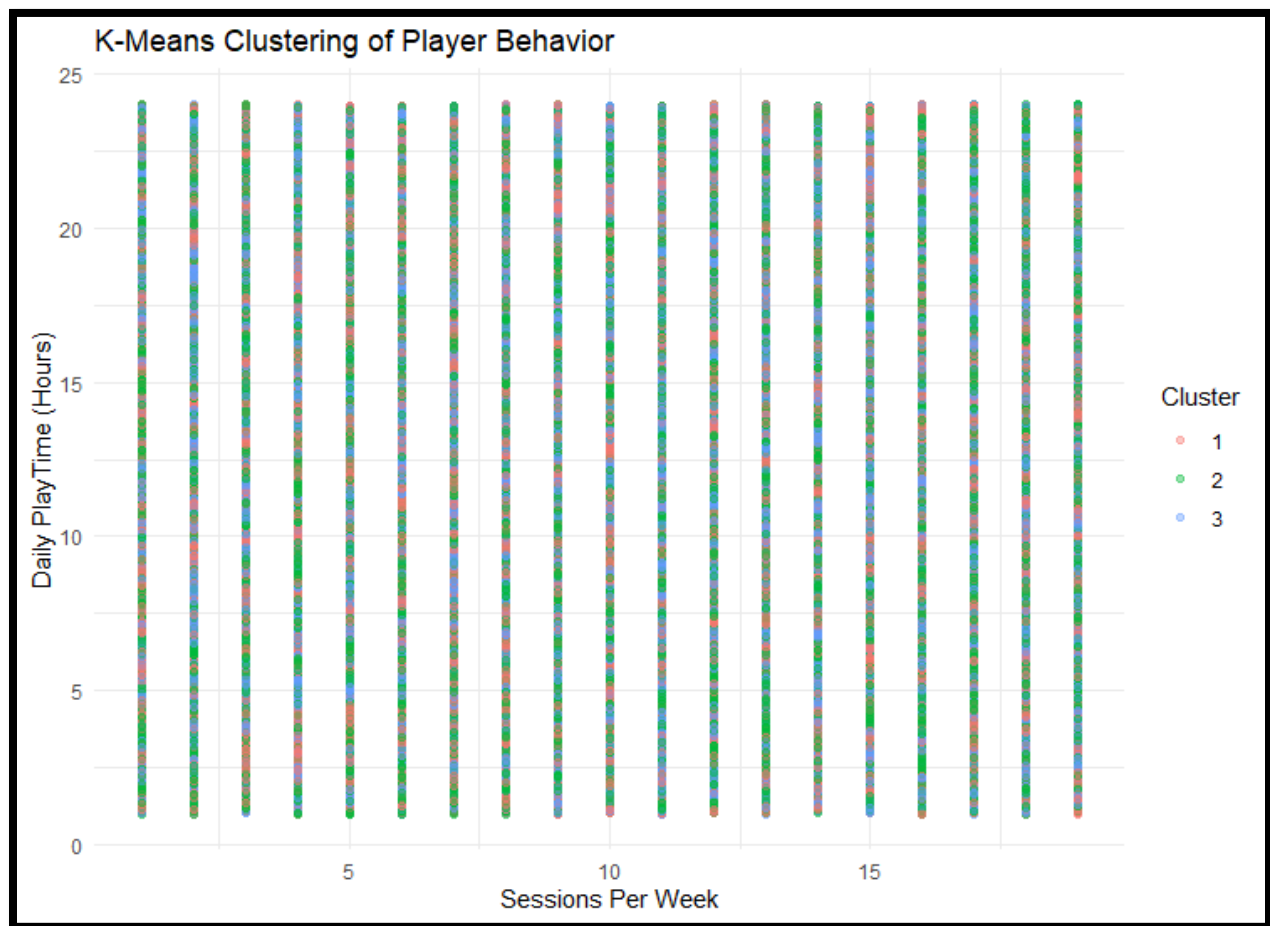
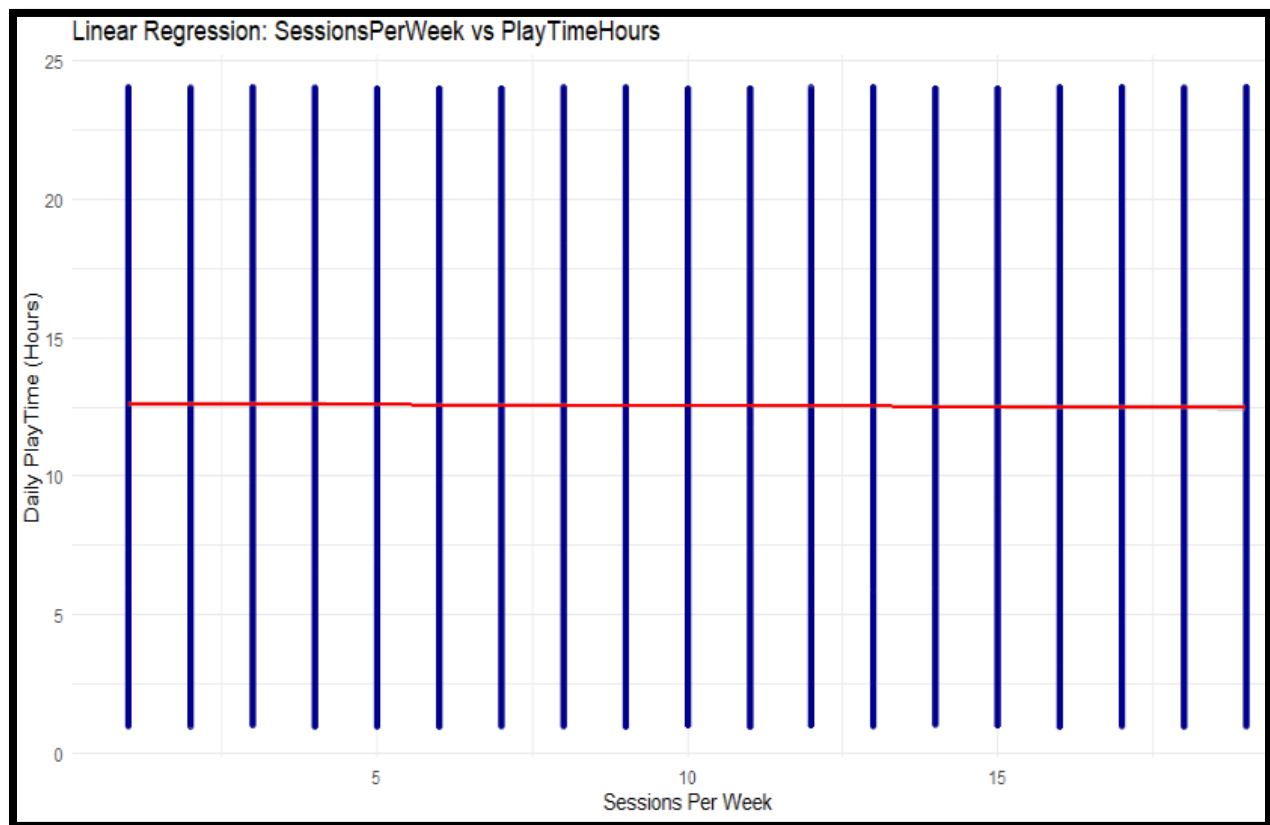


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